

feature Scaling

When my input are on different scales

Scales	example	Cgpa	Salary	Percentage
		5	19000	0
		7	29000	1
		6	54000	

Here we use Normalization or Standardization
Because while training it on these scale may
have unstable accuracy & poor neural
network, So to overcome it we use
Normalization or Standard Scaling

↓
If we know
highest & lowest
of data

↓
They both convert
in a single range
of inputs

It is better to do this every time because it
inc accuracy of Neural network

Standardization / z-Score Normalization

$$X_i \rightarrow \frac{X_i - \bar{X}}{\sigma}$$

$\bar{X} \leftarrow \text{mean}$
 $\sigma \leftarrow \text{Standard deviation}$

Age

27

15

33

63

90

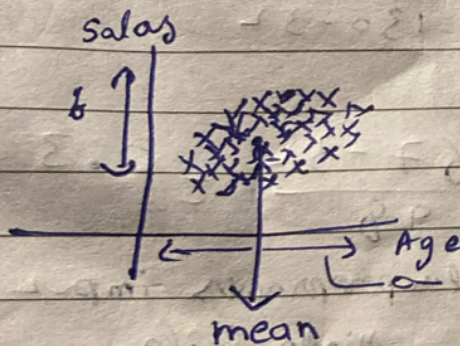
55

$$X_1 = \frac{27 - 32}{10}$$

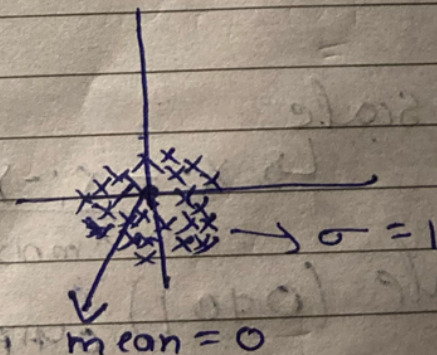
consider

$$X_2 = \frac{15 - 32}{10}$$

When all its done the new mean $\mu = 0$ &
new Standard deviation $\sigma = 1$



convert mean to 0
& $\sigma = 1$



What happen When outliers - No diff acts same
as without Standardization

Some - When to apply $\rightarrow$ K-Means

KNN

PCA

Artificial Neural Network
Gradient Descent

When not to use
Decision tree
Random Forest
k-NN
XGBoost

Normalization

- MinMax Scaling → (Widely used)
- Mean Normalization
- Max absolute Scaling
- Robust Scaling

Weights

130
67
81
32
54
⋮

Normalization - min max scal

formula in every num

find min & max values

x_{min}, x_{max}

$$x_i^0 = \frac{x_i - x_{min}}{x_{max} - x_{min}}$$

100 weights

$$x_1 = \frac{130 - 32}{130 - 32} = 1$$

$$x_2 = \frac{67 - 32}{99} = 0.3$$

MinMax Scale

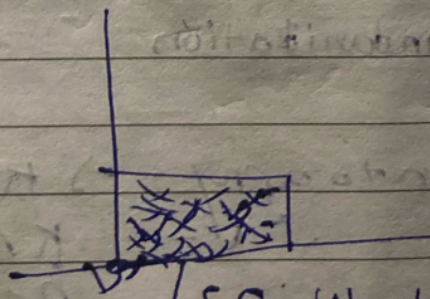
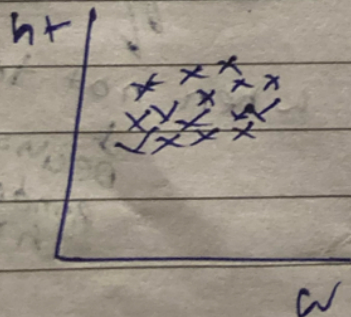
$$x_i = \frac{x_i - x_{min}}{x_{max} - x_{min}}$$

formal preprocess input

MinMax Scale

Scale = MinMax Scale (1)

→ Scale (0 to 1) output



→ Scale to 0-1

Mean Normalization

→ used for

wt norm

20

→

100

90

40

$$x' = \frac{x_i - x_{\text{mean}}}{x_{\text{max}} - x_{\text{min}}}$$

Scale $\rightarrow (-1 \text{ to } 1)$

val < mean $\rightarrow$ -ve

val > mean $\rightarrow$ +ve

No Support in Sklearn

→ we have more Alg by

useful when we need centered data

Max Abs Scaling

wt

20

100

300

$$x'_i = \frac{x_i}{|x_{\text{max}}|}$$

|x_{max}|

Scale $\rightarrow (-1, 1)$

from Sklearn.preprocessing import MaxAbsScaler

When we have sparse data

→ 0's bahot hai

Robust Scaling

wt

200

300

100

$$x_i' = \frac{x_i - x_{\text{median}}}{1.4826 (75^{\text{th}} \text{ percentile} - 25^{\text{th}} \text{ percentile})}$$

1.4826 (75th percentile - 25th percentile)

from sklearn.preprocessing import RobustScaler

Robust to outliers.

↓
good for

Normalization VS Standardization

- Is feature scaling required?

Decision tree, Random forest
XGBoost, no need

- Most of problem used in Standardization, if we have value of min max we use MinMaxScaler
Normalization

Photo

Pixel

0-255

Here we can use Normalization